(6)

ASSIGNMENT - 2

DEEP LEARNING

2. Explain how a GAN is trained.

There are 2 aspects that make generative adversarial networks more complex to train than a standard feedforward neutral network.

- The generator & the discriminator are really two neutral network which must be trained separately, but they also interact so they cannot be trained completely independently of each other.
- It is hard to identify exactly when a generative adversial network converged.
- Since the generator and discriminator have their own seperate loss functions, we have to train them separately.
- We can do this by alternating between the 2:

→We train the discriminator for one or more epochs keeping the generator weights constant.

→We train the generator for one or more epochs, keeping the discriminator weight constant.

We repeat step 1 & 2 until we determine that the network has converged.

Part 1: The discriminator is trained while the generator is idle. In this phase, the network is only forward propagated and no back propagation is done. The discriminator is trained on real data for n epochs and see if it can correctly predict them as real. Also, in this phase, the discriminator is also trained on the fake generated data from the

generator and see if it can correctly predict them as fake.

Part 2: The generator is trained while the discriminator is idle. After the discriminator is trained by the generated fake data of the generator, we can get its predicts and use the results for training the generator and get better from the previous state to try and fool the Discriminator The above method is repeated for a few epochs and then manually check the fake data if it seems genuine. If it seems acceptable, then the training is stopped, Otherwise, its allowed to continue for few more epochs.

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